

Photovoltaic Cell Parameter Analysis

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1. Overview

A solar cell converts light energy into electrical energy. Although the basic idea is simple, the electrical behavior of a solar cell depends on several physical and electrical parameters.

This project started as a simple attempt to model the current-voltage (I–V) behavior of a photovoltaic (PV) cell using Python. I first implemented an ideal single-diode model and then introduced two non-ideal effects: series resistance and shunt resistance. The main purpose of the project is to see how changing these parameters affects the I–V curve and the main performance parameters of a solar cell. The project uses NumPy, Matplotlib and SciPy for numerical calculations, plotting and solving the nonlinear equation.

The three cases studied are:

1. **Photocurrent (I_{ph})**: varied using the ideal model.
2. **Series resistance (R_s)**: varied while keeping the other parameters fixed.
3. **Shunt resistance (R_{sh})**: varied while keeping the other parameters fixed.

2. Photovoltaic (PV) cell.

A photovoltaic cell is a semiconductor device that converts incident light into electrical energy. When photons with sufficient energy strike the semiconductor, they can generate electron-hole pairs. The internal electric field of the solar cell separates these charge carriers, producing a photocurrent.

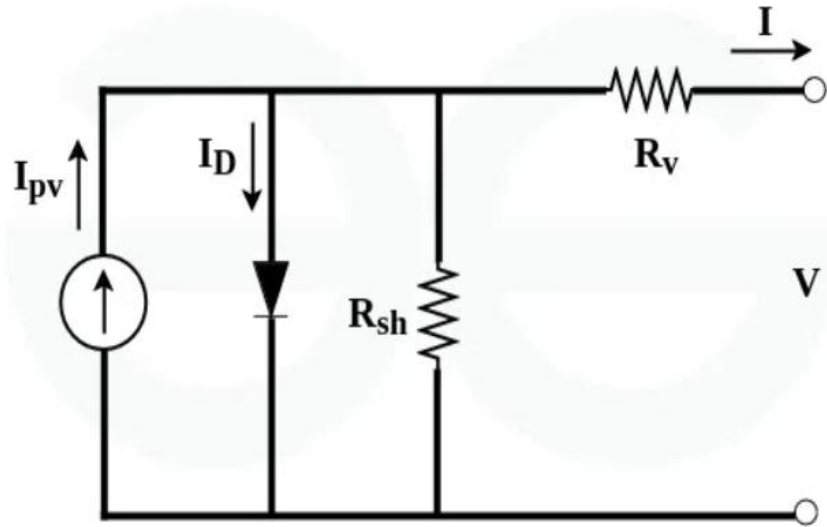
When the cell is connected to an external circuit, the separated charges can flow through the circuit and deliver electrical power.

3. Equivalent Circuit of a PV Cell

A commonly used electrical representation of a solar cell is the single-diode model (SDM). The model contains:

- A photocurrent source I_{ph}
- A diode
- Series resistance R_s
- Shunt resistance R_{sh}

A simplified representation is:



The ideal model assumes:

$R_s=0$ And, $R_{sh} \rightarrow \infty$

This removes the resistive losses and leaves the photocurrent source and diode. The more accurate or realistic single-diode model includes both resistances.

4. Single-Diode Model (SDM)

The ideal single-diode equation used in this project initially is:

$$I = I_{ph} - I_o \left(e^{\frac{q(V)}{nKT}} - 1 \right)$$

Where,

Symbols	Meaning
I	Output current
I_{ph}	Photoelectric current
I_o	Reverse saturation current
V	Terminal Voltage
n	Ideality factor
K	Boltzmann Constant
T	Absolute Temperature

The complete Single Diode Model used later is:

$$I = I_{ph} - I_0 \left(e^{\frac{q(V+IR_s)}{nKT}} - 1 \right) - \frac{V+IR_s}{R_{sh}}$$

It includes both the series and the shunt resistance. Unlike the ideal SDM equation, Current (I) appears on both sides hence it cannot be calculated directly for every voltage value. Which is later on tackled by a numeric computational method called optimization using Scipy, a python library.

5. The I–V Curve

The I–V curve shows how the output current of the solar cell changes with terminal voltage.

At approximately $V=0$, the current is close to the short-circuit current I_{sc} . As voltage increases, the current remains relatively high for a while and then begins to decrease rapidly around the knee of the curve.

Eventually, the current reaches approximately zero. This voltage is the open-circuit voltage V_{oc}

The main parameters analyzed in this project are:

Short-circuit current

$$I_{sc}=I(V=0)$$

Open-circuit voltage

$$I(V_{oc})\approx 0$$

Power

$$P=I*V$$

Maximum power

$P_{max}= I_{mp} * V_{mp}$. The voltage and current at this point are: V_{mp} and I_{mp} respectively.

The fill factor is:

$$FF= \frac{P_{max}}{V_{oc}*I_{sc}}$$

6. Project Method

The simulation was developed progressively rather than starting with the complete model immediately.

1. Ideal PV cell

I first implemented the ideal single-diode equation and generated the I–V curve by finding the current corresponding to several voltage values.

2. Change photocurrent

The photocurrent I_{ph} was varied while keeping the other parameters constant. This represents changes in the amount of light reaching the cell. Values used:

$I_{ph} = [0.030, 0.090, 0.150]$ A

The effect on I_{sc} , V_{oc} , maximum power and fill factor was then examined.

3. Add series resistance

The complete model was then used with a finite R_s . The series resistance represents internal resistance that causes voltage loss as current flows through the cell. Only R_s was varied while the other parameters were kept constant.

Values used: $[0.5, 1.5, 2.5, 3.5]$ ohm

Step 4 — Add shunt resistance

Finally, the effect of R_{sh} was investigated. Shunt resistance represents leakage paths across the cell. A lower shunt resistance allows more current to bypass the useful external circuit. Only R_{sh} was varied in this analysis.

Values used: $[50, 100, 150, 2000]$ ohm

7. Numerical Solution

When R_s and R_{sh} are included, the equation becomes implicit because the current I appears inside the exponential term. For a given voltage, the equation was rearranged into the form:

$f(I, V) = 0$. The SciPy `brentq()` function was then used to find the current numerically. The calculation is repeated for every voltage value to construct the complete I–V curve.

8. Analysis 1: Effect of Photocurrent

In the first analysis, the cell was treated as ideal that is the shunt and series resistance were ignored and only I_{ph} was changed while keeping other parameters were kept constant.

Parameter	Value
Temperature	300 K
Ideality factor	1
I_{01_0}	$1 \times 10^{-10} A$
I_{ph}	0.030, 0.090, 0.150 A
R_s	0
R_{sh}	∞

Results

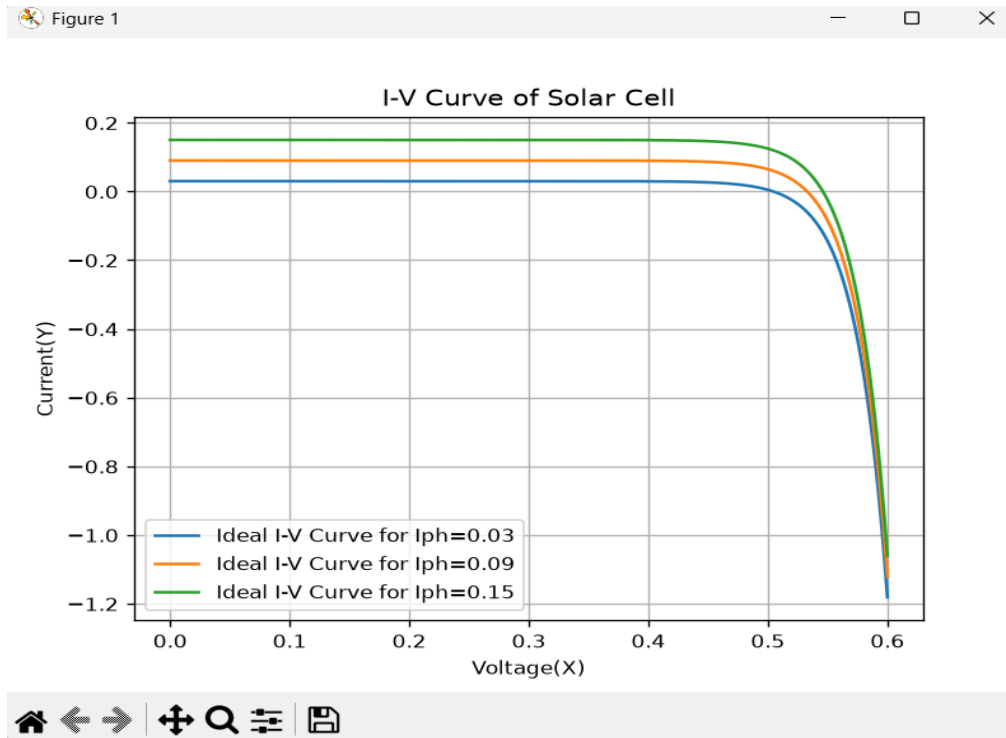
$I_{ph}(A)$	$I_{sc}(A)$	$V_{oc}(V)$	$P_{max}(W)$	$V_{mp}(A)$	$I_{mp}(A)$	FF
0.03000	0.03000	0.50450	0.01218	0.43003	0.02831	0.80448
0.09000	0.09000	0.53273	0.03894	0.45706	0.08520	0.81222
0.15000	0.15000	0.54595	0.06678	0.46967	0.14218	0.81546

Observation

Increasing I_{ph} from 0.030 A to 0.150 A (equivalent to increasing illumination from $1\times$ to $5\times$, (if 0.030 A is taken as one sun) raises I_{sc} in direct proportion, since $I_{sc} \approx I_{ph}$ when $R_s = 0$ and $R_{sh} \rightarrow \infty$. V_{oc} rises much less, from 0.5045 V to 0.5460 V, because it depends on the logarithm of the photocurrent: $V_{oc} = (nkT/q) \cdot \ln(I_{ph}/I_0 + 1)$. The data agrees with this: tripling I_{ph} adds $(nkT/q) \cdot \ln 3 \approx 28$ mV, and the table shows $0.5045 \rightarrow 0.5327$ V.

P_{max} increases from 0.01218 W to 0.06678 W ($5.5\times$ for $5\times$ the photocurrent). This is slightly faster than linear because $P_{max} = V_{oc} \cdot I_{sc} \cdot FF$, and V_{oc} and FF also rise. The extra power comes mainly from current: I_{mp} grows almost in proportion to I_{ph} ($0.0283 \rightarrow 0.1422$ A) while V_{mp} changes little ($0.430 \rightarrow 0.470$ V). The fill factor rises modestly ($0.804 \rightarrow 0.815$) because a higher V_{oc} , relative to the thermal voltage nkT/q , gives a sharper knee on the I–V curve.

I-V curve



9. Analysis 2: Effect of Series Resistance

We introduce both the series and the shunt resistance but for this case, we only vary series resistance while keeping the shunt resistance and other variables constant.

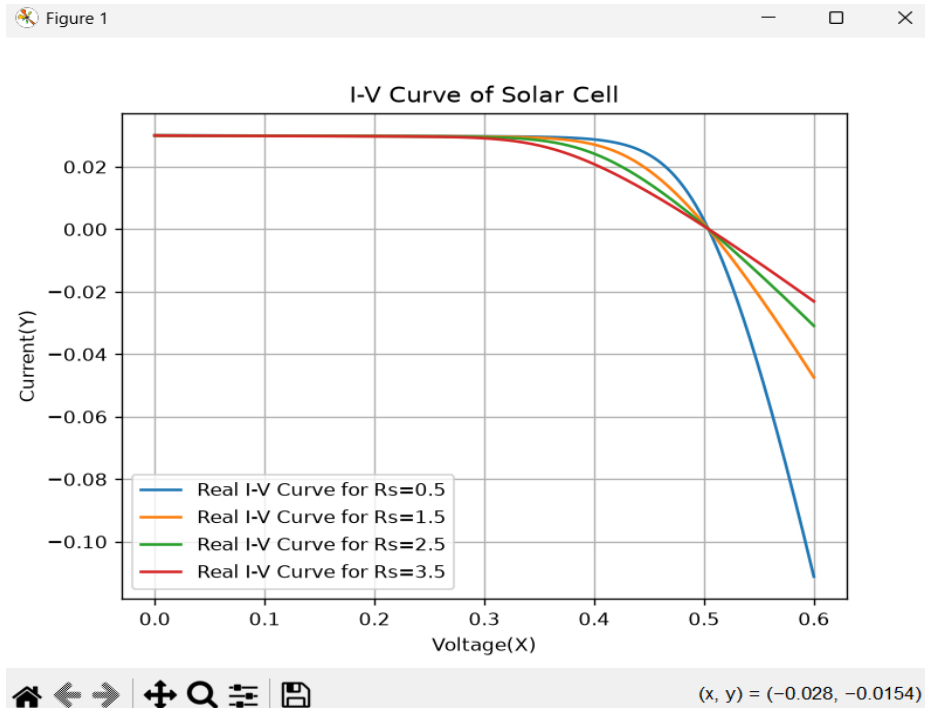
Parameter	Value
Temperature	300 K
Ideality factor	1
I_o	$1 \times 10^{-10} A$
I_{ph}	0.030 A

Parameter	Value
Rs	0.5, 1.5, 2.5, 3.5 ohm
Rsh	1000 ohm

Results

Rs(ohm)	Isc(A)	Voc(V)	Pmax(W)	Vmp(A)	Imp(A)	FF
0.50000	0.02999	0.50390	0.01160	0.41742	0.02780	0.76792
1.50000	0.02996	0.50390	0.01084	0.39279	0.02759	0.71788
2.50000	0.02993	0.50390	0.01008	0.36997	0.02726	0.66875
3.50000	0.02990	0.50390	0.00935	0.34835	0.02685	0.62080

I-V curve



Observation

Increasing R_s from 0.5Ω to 3.5Ω reduces P_{max} from 0.01160 W to 0.00935 W (a 19% loss) and the fill factor from 0.768 to 0.621 . The cause is the internal voltage drop $I \cdot R_s$: at any current, the terminal voltage is lower than the voltage across the junction, so the knee of the curve rounds off and V_{mp} falls ($0.417 \rightarrow 0.348 \text{ V}$). I_{mp} also falls slightly ($0.0278 \rightarrow 0.0269 \text{ A}$).

V_{oc} stays fixed at 0.5039 V for every R_s value. At open circuit $I = 0$, so there is no drop across R_s , and this is why all four curves cross at the same point. I_{sc} decreases only marginally ($0.02999 \rightarrow 0.02990 \text{ A}$) because at $V = 0$ the voltage $I \cdot R_s$ appears across R_{sh} and diverts a small leakage current.

10. Analysis 3: Effect of Shunt Resistance

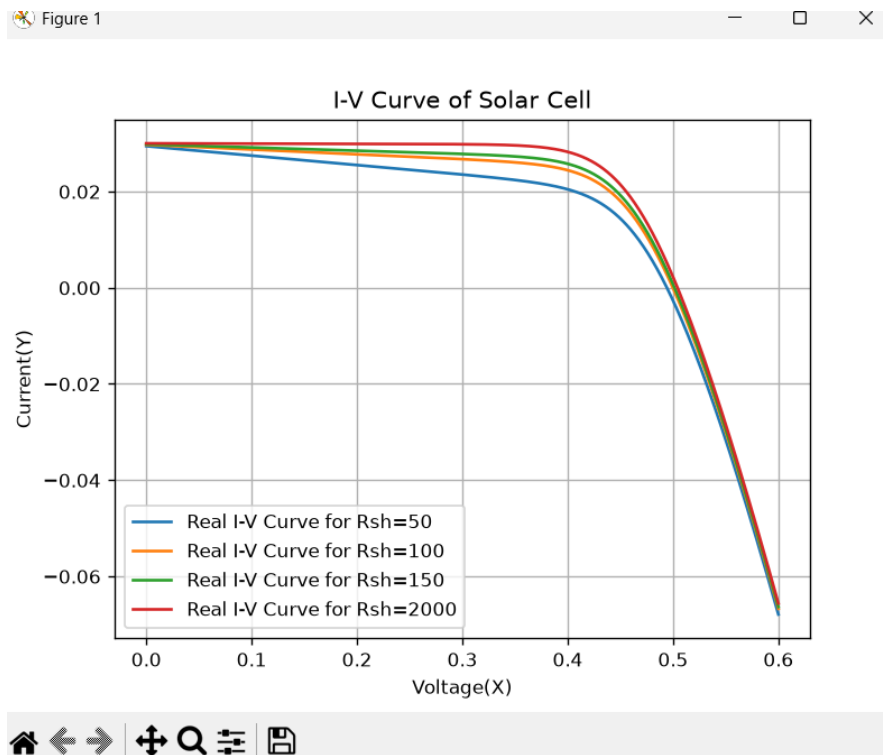
Furthermore, we vary the shunt resistance while keeping the series resistance and other parameters constant and analyze the data and the IV curve.

Parameter	Value
Temperature	300 K
Ideality factor	1
I_0	$1 \times 10^{-10} A$
I_{ph}	0.030 A
R_s	1 ohm
R_{sh}	50, 100, 150, 2000 ohm

Results

$R_{sh}(\text{ohm})$	$I_{sc}(A)$	$V_{oc}(V)$	$P_{max}(W)$	$V_{mp}(A)$	$I_{mp}(A)$	FF
50.00000	0.02941	0.49429	0.00818	0.39399	0.02077	0.56291
100.00000	0.02970	0.49970	0.00976	0.40120	0.02433	0.65761
150.00000	0.02980	0.50150	0.01030	0.40300	0.02555	0.68898
2000.00000	0.02999	0.50450	0.01130	0.40541	0.02787	0.74694

I-V curve



Observation

Increasing R_{sh} from 50Ω to 2000Ω improves every performance parameter: P_{max} rises from 0.00818 W to 0.01130 W (+38%) and the fill factor from 0.563 to 0.747 . V_{oc} rises from 0.4943 V to 0.5045 V , I_{sc} from 0.02941 A to 0.02999 A , and V_{mp} and I_{mp} also increase. In other words, a low R_{sh} lowers all of these, so leakage reduces both current and voltage.

The reason is the leakage current V/R_{sh} , which grows linearly with voltage. This is why the low- R_{sh} curves slope downward instead of staying flat, with a slope near $V = 0$ of about $-1/R_{sh}$. At the maximum power point with $R_{sh} = 50 \Omega$, about $0.394/50 \approx 8 \text{ mA}$ leaks through the shunt path, roughly a quarter of the 30 mA photocurrent. With $R_{sh} = 2000 \Omega$ the leakage is only about 0.2 mA , under 1%.

11. Comparison of the Three Parameters

The three analyses show that the parameters in the single-diode model affect different parts of the photovoltaic cell's I–V characteristics.

Effect of Photocurrent I_{ph}

Increasing I_{ph} increases the current generated by the cell. This is mainly reflected in a higher short-circuit current I_{sc} . The open-circuit voltage also increases, but at a slower rate because its relationship with photocurrent is logarithmic. Therefore, changing I_{ph} mainly changes the current level and the amount of power that can be produced.

Effect of Series Resistance R_s

Increasing R_s introduces greater internal voltage loss:

$V_{Rs} = I R_s$: As a result, the I–V curve becomes more distorted around the knee region. The maximum power and fill factor decrease as R_s increases. In the model used here, V_{oc} remains approximately unchanged because the current is zero at open circuit, making the voltage drop across R_s zero.

Effect of Shunt Resistance R_{sh}

The shunt resistance controls the amount of current that can leak through the cell rather than being delivered to the external circuit. A lower R_{sh} produces greater leakage and causes the current to fall more rapidly with increasing voltage. As R_{sh} becomes larger, the effect of this leakage becomes smaller and the curve approaches the ideal behavior.

Overall Comparison

Parameter	Physical meaning	Main effect on I–V curve
I_{ph}	Light-generated current	Changes the overall current level
R_s	Internal series resistance	Distorts the knee and reduces useful power
R_{sh}	Leakage resistance	Causes additional current loss

These simulations show why the ideal PV model is useful for understanding the basic behavior of a solar cell, but also why resistive losses need to be considered when trying to represent a more realistic device.

12. Conclusion

This project developed a numerical model of a photovoltaic cell using the single-diode model and Python. I started with the ideal case and then gradually introduced series and shunt resistance to investigate how different parameters affect the electrical behavior of the cell. The ideal model showed the relationship between photocurrent and the I–V characteristics of the cell. Increasing the photocurrent increased the current produced by the cell and consequently increased the available power. The addition of series resistance showed how internal voltage losses can reduce the maximum power and fill factor, while the shunt-resistance analysis demonstrated the effect of leakage current on the I–V curve.

An important part of the project was that the complete single-diode equation could not be solved directly for current because current appears inside the exponential term. I therefore used numerical root finding with SciPy method to calculate the current at each voltage point.

Overall, the project provided a practical way to connect the basic physics of photovoltaic cells with numerical modeling and data analysis. Instead of only studying the equations, I was able to change individual parameters, observe the resulting curves, and quantify their effects through I_{sc} , V_{oc} , P_{max} , V_{mp} , I_{mp} and fill factor.

13. Future Improvements

The current project uses simulated parameters, so there are several ways it could be developed further.

1. Include temperature variation

Temperature has a significant effect on PV cell behavior. The model could be run at several temperatures to study changes in V_{oc} , I_{sc} , and maximum power.

2. Compare simulation with experimental data

A physical solar cell could be tested under controlled conditions and its measured I–V data could be compared with the simulated curve. This would allow the model parameters to be adjusted based on real measurements rather than assumed values.

3. Parameter extraction

Instead of choosing I_{ph} , I_o , R_s , R_{sh} , and n beforehand, measured I–V data could be used to estimate these parameters. This would move the project from simply simulating a PV cell toward characterizing an actual device.

4. Study multiple parameters together

The current analysis changes one parameter at a time. A later version could investigate combinations such as temperature and irradiance, or the interaction between R_s and R_{sh} .

An analytical project by Ritiz Shrestha